

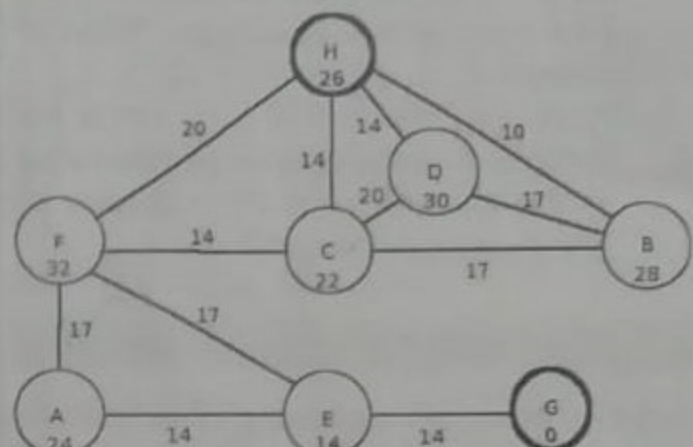
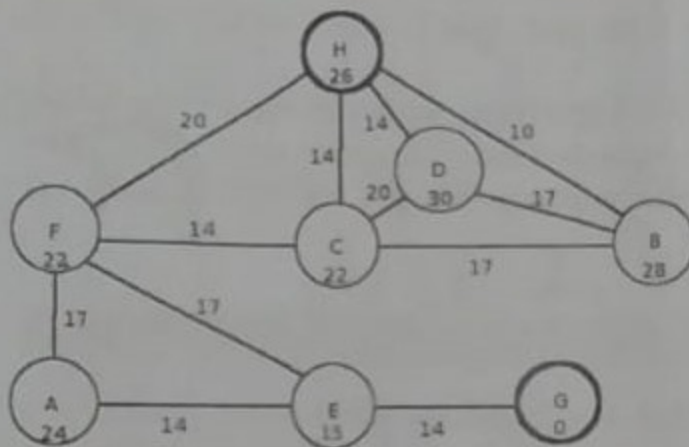


Exercise I (Heuristics)

(20pts)

Suppose you are asked to find the shortest path from H to G in the graphs below. For both of the graphs answer the following questions:

1. Is the heuristic function admissible? Explain
2. Is the heuristic function Consistent? Explain

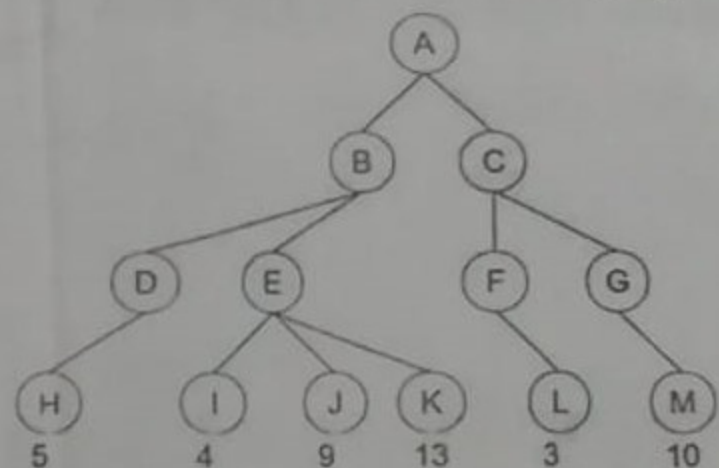


3. Apply the algorithms uniform cost, breadth first and best first on the first graph.

Exercise II (Games)

(25pts)

1. Perform Minimax on this tree.
2. Tree Rotation and Alpha-Beta Pruning
 - a) Using the minimax calculations from part 1, Perform the alpha-beta pruning on the same tree.



- b) Rotate the children of each node in the above tree at every level to ensure maximum alpha-beta pruning. Fill in the nodes with the letter of the corresponding node. Draw the new edges.

MAX

A

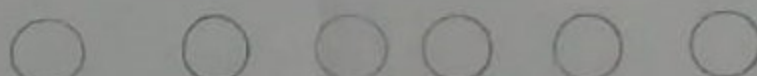
MIN



MAX



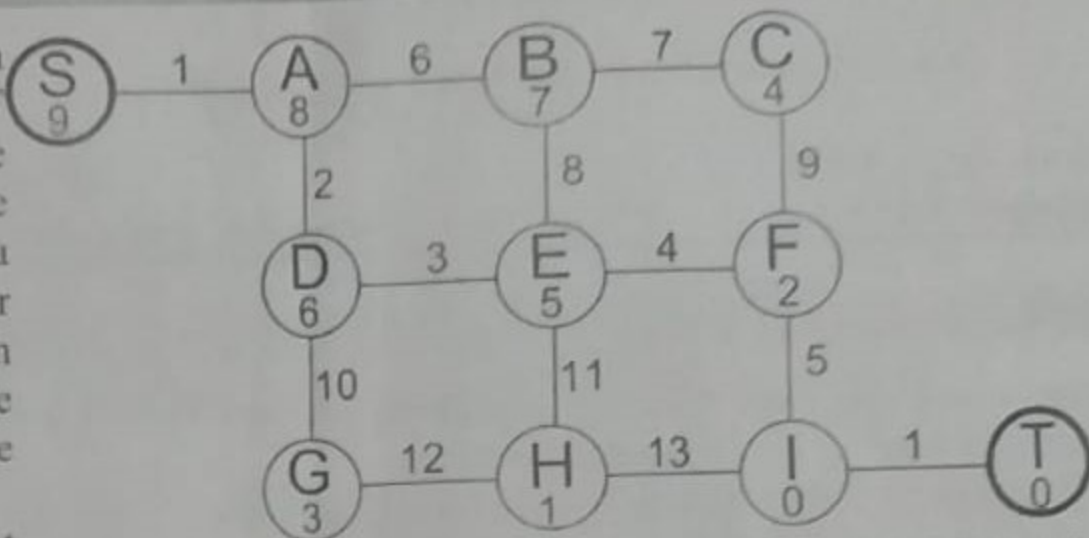
MIN



(15pts)

Exercise III (Search)

Consider the following search graph. S is the start node and T is the goal node. The edges are labeled with distances, and the nodes are labeled with a heuristic estimate to your destination at T. When performing search, ties are broken by choosing the node that is **alphabetically first**.



1. Use a **Depth-first search with an expanded list**. Draw your search tree. Show your expanded list at the time your search terminates. What is the final path that is found using depth-first search?
2. Next, you try to find your route using **A* algorithm with the heuristic values indicated on the graph and also using an expanded list**. Draw your search tree. Show your expanded list at the time your search terminates. What is the final path that is found using A* search?

Exercise IV (Problem Formulation)

(15pts)

You have three jugs measuring 12 gallons, 8 gallons, and 3 gallons, and a water faucet. You can fill the jugs up from the faucet, or empty them out from one to another or onto the ground. You need to measure out exactly 1 gallon.

Give a complete problem formulation for the above given problem. Choose a formulation that is precise enough to be implemented.